

1 The Frontier-Defect Conservation Theorem

Let $\mathcal{K}_n$ denote the set of root laws of exact depth- n zero-repair trees and let

$$\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}B, \quad \mathcal{B}_0 := 0.$$

Throughout this section we only use

$$\mathbb{E}P = \mathbb{E}Q.$$

Let

$$\{B_h\}_{|h| \leq n}$$

be an exact depth- n zero-repair tree, so that

$$Q * B_h = \sum_x P(x) \delta_x * B_{hx}.$$

For a history h , write

$$r(h) := n - |h|$$

for the remaining horizon and define its *frontier defect*

$$\Delta_h := \mathbb{E}B_h - \mathcal{B}_{r(h)}.$$

Since

$$B_h \in \mathcal{K}_{r(h)},$$

one always has

$$\Delta_h \geq 0.$$

For a word

$$g = (g_1, \dots, g_s) \in \mathcal{A}^s,$$

write

$$P(g) := \prod_{j=1}^s P(g_j).$$

Theorem 1 (Frontier-defect conservation). *Let h be any node of an exact zero-repair tree with*

$$r := r(h).$$

Then for every

$$0 \leq s \leq r,$$

one has the exact identity

$$\sum_{g \in \mathcal{A}^s} P(g) \Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}. \quad (1)$$

In particular, for one generation,

$$\sum_x P(x) \Delta_{hx} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-1}. \quad (2)$$

If h lies on the optimal frontier,

$$\Delta_h = 0,$$

then

$$\sum_{g \in \mathcal{A}^s} P(g) \Delta_{hg} = \mathcal{B}_r - \mathcal{B}_{r-s}. \quad (3)$$

Proof. Taking expectations in the exact zero-repair identity gives

$$\mathbb{E}Q + \mathbb{E}B_h = \mathbb{E}P + \sum_x P(x)\mathbb{E}B_{hx}.$$

Since

$$\mathbb{E}P = \mathbb{E}Q,$$

we obtain

$$\mathbb{E}B_h = \sum_x P(x)\mathbb{E}B_{hx}. \quad (4)$$

Iterating (4) for s generations yields

$$\mathbb{E}B_h = \sum_{g \in \mathcal{A}^s} P(g)\mathbb{E}B_{hg}. \quad (5)$$

Every descendant hg has remaining horizon $r - s$, hence

$$\mathbb{E}B_{hg} = \mathcal{B}_{r-s} + \Delta_{hg}.$$

Substituting into (5),

$$\mathbb{E}B_h = \mathcal{B}_{r-s} + \sum_{g \in \mathcal{A}^s} P(g)\Delta_{hg}.$$

On the other hand,

$$\mathbb{E}B_h = \mathcal{B}_r + \Delta_h.$$

Therefore

$$\mathcal{B}_r + \Delta_h = \mathcal{B}_{r-s} + \sum_g P(g)\Delta_{hg},$$

which rearranges to

$$\sum_g P(g)\Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.$$

This proves (1). The remaining statements are special cases. $\square$

1.1 Counterfactual conservation

A node h will be called a *contact node* if

$$\Delta_h = 0,$$

equivalently

$$\mathbb{E}B_h = \mathcal{B}_{r(h)}.$$

Thus the subtree rooted at h is itself optimal for its remaining horizon.

The preceding theorem has an immediate consequence.

[Counterfactual bill decomposition] Let h be a contact node with remaining horizon r , and suppose that for some word

$$g_* \in \mathcal{A}^s$$

the descendant hg_* is again a contact node.

Then

$$\boxed{\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{\substack{g \in \mathcal{A}^s \\ g \neq g_*}} P(g)\Delta_{hg}.} \quad (6)$$

Hence the realised contact history g_* contributes exactly zero to the excess bill, while the entire increase

$$\mathcal{B}_r - \mathcal{B}_{r-s}$$

is carried by the alternative histories.

Proof. By Theorem 1,

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_g P(g) \Delta_{hg}.$$

Since hg_* is a contact,

$$\Delta_{hg_*} = 0.$$

Its term can therefore be removed from the sum. $\square$

This gives an exact mathematical meaning to the distinction between a realised history and the ensemble of unrealised futures:

the realised optimal branch may carry no frontier defect at all,

while

the full cost of extending the horizon is stored in the counterfactual branches.

1.2 Scale-by-scale form

Define the marginal horizon cost

$$d_r := \mathcal{B}_r - \mathcal{B}_{r-1}.$$

[Scale-by-scale export law] Suppose

$$h_0 \prec h_1 \prec \dots \prec h_\ell$$

is a sequence of contact nodes, where

$$h_{j+1} = h_j x_{j+1}.$$

If the remaining horizon at h_0 is r , then for every $0 \leq j < \ell$,

$$d_{r-j} = \sum_{x \neq x_{j+1}} P(x) \Delta_{h_j x}. \tag{7}$$

Thus every additional unit of horizon creates exactly one scalar marginal cost d_{r-j} , and this cost is exported into the sibling futures whenever the selected branch remains on the optimal frontier.

Proof. Apply (2) at h_j . Since both h_j and $h_j x_{j+1}$ are contact nodes,

$$\Delta_{h_j} = 0, \quad \Delta_{h_j x_{j+1}} = 0.$$

Hence

$$d_{r-j} = \sum_x P(x) \Delta_{h_j x} = \sum_{x \neq x_{j+1}} P(x) \Delta_{h_j x}.$$

$\square$

1.3 The branching martingale

Let

$$H_t = (X_1, \dots, X_t), \quad X_1, X_2, \dots \stackrel{\text{iid}}{\sim} P,$$

be a random history.

[Renormalised frontier martingale] For an optimal depth- n tree,

$$\boxed{Z_t := \Delta_{H_t} - (\mathcal{B}_n - \mathcal{B}_{n-t})} \tag{8}$$

is a martingale.

Equivalently,

$$\boxed{\mathbb{E}\Delta_{H_t} = \mathcal{B}_n - \mathcal{B}_{n-t}.$$

Proof. The one-step identity gives

$$\mathbb{E}[\Delta_{H_{t+1}} \mid H_t] = \Delta_{H_t} + d_{n-t}.$$

Since

$$\mathcal{B}_n - \mathcal{B}_{n-t-1} = \mathcal{B}_n - \mathcal{B}_{n-t} + d_{n-t},$$

subtracting the deterministic frontier term proves

$$\mathbb{E}[Z_{t+1} \mid H_t] = Z_t.$$

At the root,

$$Z_0 = 0,$$

because the tree is optimal. □

1.4 Interpretation

The theorem rules out the naive interpretation in which every branch creates an independent additional bill.

At depth s below a frontier contact there may be 5^s possible histories, but their weighted frontier defects satisfy the single exact constraint

$$\boxed{\sum_{g \in \mathcal{A}^s} P(g) \Delta_{hg} = \mathcal{B}_r - \mathcal{B}_{r-s}.$$

Thus the exponentially many branches are not associated with exponentially many independent costs.

They are a decomposition of one conserved scalar defect.

If one descendant remains on the optimal frontier, then that realised history carries none of the defect:

$$\Delta_{hg_*} = 0,$$

and all of the cost resides in histories which did not occur.

In this precise sense,

$$\boxed{\text{the cost belongs to the ensemble of possible futures, not to the realised history alone.}}$$

The recursive appearance of lower-horizon optimal laws inside larger computed optimisers is therefore not merely a visual self-similarity. It is the equality case of the exact conservation law (6).

1.5 What this theorem closes

The structural statement is now complete:

An exact zero-repair tree carries a conserved frontier defect. At every scale, the increase in optimal bill is redistributed among the possible future branches. Whenever a realised branch reproduces a lower-horizon optimum, its frontier defect is exactly zero and the entire marginal cost is necessarily carried by its counterfactual siblings.

 (9)

No finite-memory assumption, common-anchor conjecture, asymptotic fit, or uniqueness of the optimiser is used.